

Project 302: System Audit Report

Environment Snapshot (Live): Docker ai_tutoring_pg healthy on 5433 | Ollama serving qwen3:8b, qwen3.5:9b, deepseek-r1:14b | Python 3.14.3 | Node 24 | Docker 29.8

Repository: github.com/PatrickLuhanga/AI-tutoring-system.git (clean tree)

1. Present & Functional

Component	Status	Evidence in Codebase
PostgreSQL 16 + pgvector	WORKING	docker-compose.yml healthy on 5433; 9 tables + HNSW indexes in src/models.py
Relational Store	MATCHES SPEC	src/models.py — CHECK constraints, unique keys, JSONB flag arrays setup for RBAC, telemetry, feedback.
RAG Pipeline (§8)	MATCHES SPEC	Module-filtered cosine search applied before similarity (src/retriever.py:141); 4-module registry active.
Flask API Gateway	WORKING	src/app.py factory active; endpoints /api/chat, /api/feedback, /api/modules functional with clean-JSON handlers.
Agent Workflow Layer	WORKING	Separation of concerns met. Intent→Scaffolding→Retriever→Tutor→Guardrail coordinated in workflow.py.
Guardrail Agent (§11)	IMPLEMENTED	Code-leak/direct-answer/out-of-scope checks working. Block/truncate + Socratic fallback active (src/agents/guardrail.py).
Dynamic LLM Router	EXCEEDS SPEC	Runtime local↔cloud switch via POST /api/admin/llm-config; Fernet-encrypted keys active (src/llm_router.py).
Inference Fault Isolation	WORKING	Timeout + thread-safe circuit breaker active (src/inference/ollama_client.py).
React UI (Client Tier)	WORKING	Vite + React 19 + Tailwind 4 functional. StudentChat with module dropdown, session, audit panel, thumbs feedback.
Socratic Progression	WORKING	5-stage sequence + bypass pinning active (scaffolding.py:26).

2. Present but Needs Work (Refinements Required)

- **Intent Agent Rigidity:** English-only regex heuristics with no tolerance for typos/slang. Misroutes "difference between" to factual instead of conceptual. Silently falls back to heuristics despite LLM path existing.
- **Socratic Prompting Constraints:** Stage advances purely by turn count, not by student quality. Every bypass is pinned to the same hint. Prompts are hardcoded strings without an admin-tuning mechanism. Few-shot examples are placeholders, not real corpus excerpts.
- **Telemetry Logging Gaps:** Failed inference turns are never logged (exception raised before DB interceptor). Student-message/reply text is not stored. "Abandoned sessions" and "turns-to-resolution" metrics cannot be computed.
- **Authentication & RBAC:** Auth is a stub. Strict DUT4life mode raises 501. No endpoints exist to grant tutor privileges or manage tutor assignments, even though the database tables are ready.
- **Admin-Key Mismatch Bug:** Frontend uses `VITE_ADMIN_KEY=dev-admin-key` while backend expects `ADMIN_API_KEY=change-me-admin-key`, resulting in 401 Unauthorized errors on the live dashboard.
- **Model Drift & Embeddings:** Docs specify `qwen3:4b` but `qwen3:8b` is in use. Embedding model deviates from architecture (using `all-MiniLM-L6-v2` instead of `nomic-embed-text`).
- **Guardrail Heuristics:** Relies on weak proxies (regex + lexical overlap 0.08) for out-of-scope detection rather than a semantic LLM audit pass. Thresholds remain unevaluated.
- **Frontend Integration:** The module dropdown uses hardcoded mock data, ignoring the functional `GET /api/modules` backend route.
- **Testing:** Zero test files exist. The accuracy/precision/recall/F1 evaluation harness defined in the architecture paper is absent.

3. Missing Entirely

- **MISSING Admin Analytics Route:** Frontend attempts to call `GET /api/admin/analytics`, but no corresponding backend route exists to serve these metrics.
- **MISSING Scoped Tutor Dashboard:** Only a combined Admin/Tutor view exists. Database-level module scoping for tutors is unsupported by the backend.
- **MISSING Admin RBAC Management:** No endpoints to grant tutor privileges, manage assignments/enrollments, or seed the student directory.
- **MISSING DUT4life Integration:** Microsoft identity-provider integration and login UI are completely absent.
- **MISSING Evaluation & Maintenance:** No evaluation harness (test set + ground-truth). No prompt-adjustment cycle (prompts are locked Python constants). Code-repair maintenance workflow undefined.
- **MISSING Documentation Sync:** The updated architecture PDF is untracked in Git. A fresh clone will not receive the latest specifications.

4. Developer Onboarding (Quick Start)

To get a new teammate up and running, OpenCode has generated a `GETTING_STARTED.md` file in the repository root. Below is a summary of the required setup commands.

1. Clone and Configure Environment

```
git clone https://github.com/PatrickLuhanga/AI-tutoring-system.git cd AI-tutoring-system # Backend .env copy .env.example .env # Important: Ensure ADMIN_API_KEY matches the frontend key # Frontend .env cd frontend copy .env.example .env cd ..
```

2. Install Dependencies

```
# Backend (Windows) python -m venv venv .\venv\Scripts\activate pip install -r requirements.txt # Frontend cd frontend npm install cd ..
```

3. Boot Infrastructure

```
# Start PostgreSQL + pgvector (Terminal 1) docker-compose up -d # Verify Database on port 5433 docker ps # Start LLM Runtime (Terminal 2) ollama run qwen3:8b
```

4. Launch Microservices

```
# Terminal 3: API Gateway python app.py # Terminal 4: React UI cd frontend npm run dev
```